Individual Signatures of Spillover in Naturalistic Reading

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Background: Spillover effect happens when the processing difficulty of one word affects reading times on the following, but its causes and stability across individuals are still unknown. Classic reading models, such as E-Z Reader and SWIFT, link spillover to oculomotor planning or parallel processing, assuming it remains stable across individuals. Sentence processing models, e.g., surprisal theory, often treat it as a nuisance. Some recent studies suggest that spillover may reflect deeper cognitive processes, but it is still unclear whether it reflects a stable reader trait, a temporary state, or an interaction of both. This project investigates the stability of individual spillover effects across reading contexts, examines their modulation by cognitive profiles, and tests whether they are associated with stable reading behaviors.

Method: We used six eye-tracking datasets that allow us to compare each participant's spillover effects across different reading contexts. In InDiCo (German) [1], participants read in two sessions, with a two-week interval between them. In PoTeC (German) [2], participants read both within and outside their expertise domain. In OneStop (English) [3], participants did both normal naturalistic reading and repeated reading. In GECO-NL [4] and GECO-ZH [5], Dutch and Chinese participants read in both their native language (L1) and English (L2). In HKC (Chinese) [6], participants read both single isolated sentences and full paragraphs. For each participant, we estimated spillover effects of word length, lexical frequency, surprisal, and their stability using a Bayesian two-task hierarchical model [7]. We then tested how these spillover effects are modulated by psychometric profiles and related to skipping and regression behaviors.

Results: Spillover effects showed different levels of stability across reading contexts **(2nd row in Fig. 1)**. Word length spillover was the most stable, while frequency and surprisal spillover were less consistent. People showed different spillover patterns depending on the reading contexts—they did not adopt the same strategy across situations. Spillover effects were more stable across sessions, domains, and when the L1 and L2 were similar (e.g., Dutch-English), but less stable across distant languages (e.g., Chinese-English), reading regimes, and context lengths. Spillover effects are not uniform across readers, and are modulated by reading fluency **(Fig 2)**. Low-fluency readers showed resource-driven spillover, with stronger effects of low-level difficulty such as word length and frequency. In contrast, high-fluency readers showed integrative spillover, driven mainly by surprisal. These individual spillover signatures were systematically related to reading behavior: greater sensitivity to previous word length and frequency was associated with less skipping and more regressions, while greater sensitivity to previous word surprisal was associated with more skipping and fewer regressions.

Discussion: Our results show that spillover is neither noise nor a fixed processing artifact. Spillover reflects both stable reader-specific traits and context-sensitive adaptations. The contrasting patterns observed for word length and frequency versus surprisal spillover, in their stability, cognitive modulation, and correlations with other reading behaviour, suggest distinct underlying mechanisms: low-level resource limitations versus higher-level integrative processing. These results highlight the importance of individual difference approaches for understanding spillover and challenge population-level analyses that obscure meaningful heterogeneity in reading.

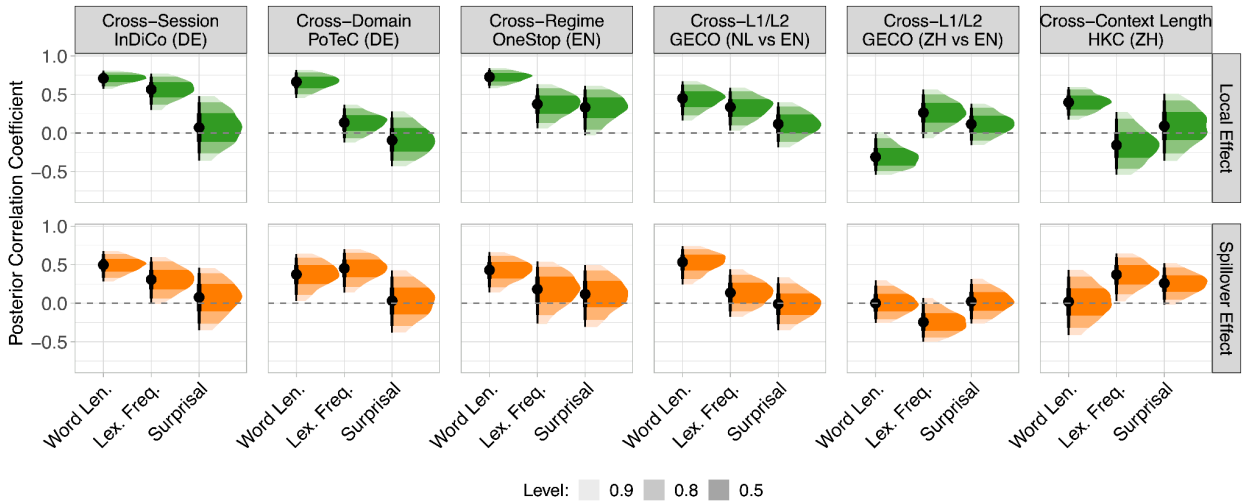

Fig. 1: **Measurement reliability of individual spillover effects**: the second row shows the posterior distributions of cross-context correlation coefficients (reliability estimates) of previous word length, lexical frequency, and surprisal effects on gaze duration on the current word. Reliability is assessed across sessions (InDiCo), expertise domains (PoTeC), reading regimes (OneStop), L1 vs L2 (GECO-NL, GECO-ZH), and context length (HKC). The first row shows the reliability estimates for the immediate local words, for comparison.

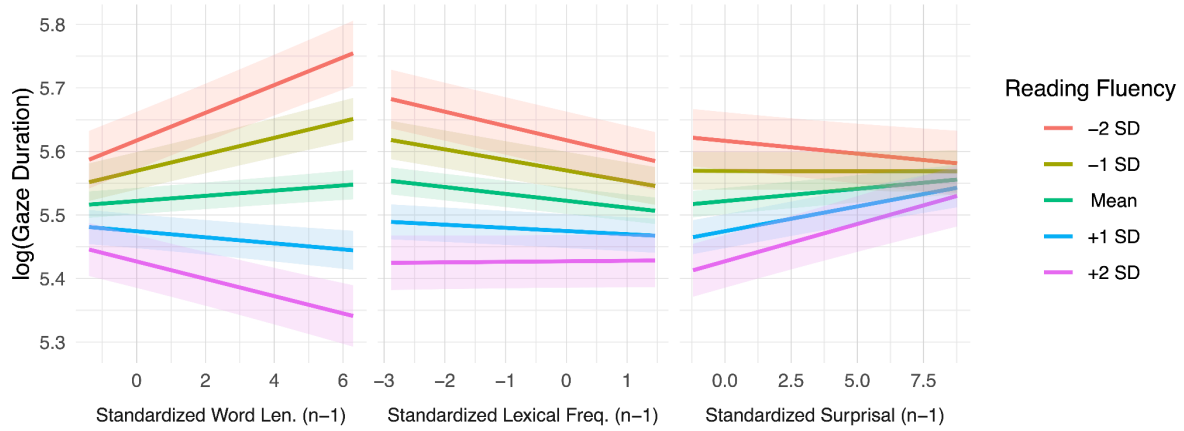

Fig. 2: Reading fluency modulates the spillover of word length, lexical frequency, and surprisal differently.

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